

KUMAR VIBHANSHU

Technical Lead | Cloud & DevOps Engineer | Microservices & AI Solutions

Gurugram, Haryana, India | 8809233239 | kmr vibhanshu@gmail.com | linkedin.com/in/kumar-vibhanshu | github.com/kumar-vibhanshu

PROFESSIONAL SUMMARY

Technical Lead with 5.5+ years of software engineering experience, building scalable microservices and distributed systems while architecting secure, multi-cloud infrastructure on AWS and Azure using Kubernetes, Docker, Terraform, and CI/CD pipelines. Proven track record of improving system reliability and delivery efficiency, including reducing deployment time by 40%. AWS Certified Solutions Architect and Azure AI Engineer Associate, experienced in leading cross-functional teams with strong client-facing collaboration.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, Lambda, VPC, ECR, EKS, RDS, CloudFormation, IAM, SQS, SNS, CloudFront, ElastiCache), Microsoft Azure (App Services, Functions, Storage)

DevOps & Automation: GitHub Actions, Jenkins, CI/CD Pipelines, Docker, Kubernetes, Terraform, Infrastructure as Code, Git, Maven

Observability & Monitoring: CloudWatch, Prometheus, Grafana, New Relic, Kibana

Architecture: Microservices, RESTful APIs, API Gateway, Event-Driven Architecture, Serverless Architecture, Circuit Breaker & Resilience Patterns, Reactive & Async Workflows

AI & Machine Learning: Azure OpenAI Service, Azure Cognitive Services, Generative AI, MCP Servers, Agentic AI

Messaging & Streaming: AWS SQS/SNS, Pub/Sub, Apache Kafka

Databases: PostgreSQL, MySQL, SQL Server, Redis

Tools & Practices: Postman, Bruno, JIRA, Confluence, SonarQube, Checkmarx, Agile/Scrum

PROFESSIONAL EXPERIENCE

Technical Lead

Feb 2025 - Present

Tata Consultancy Services

Gurugram, Haryana, India

- Lead technical teams in designing and delivering enterprise-grade cloud and AI solutions for high-profile clients.
- Architect multi-cloud solutions leveraging AWS and Azure services to support millions of users with high availability.
- Implement AI-powered applications using Azure OpenAI Service and Azure Cognitive Services to enable intelligent automation.
- Drive DevOps best practices and automate CI/CD pipelines using GitHub Actions, reducing deployment time by 40%.
- Mentor development teams on cloud architecture patterns, microservices design, and AI implementation strategies.
- Rebuilt a commerce platform with an agentic AI experience, building MCP servers and backend APIs that enable AI agents to handle product lookup, pricing estimation, and order placement.
- Own end-to-end cloud infrastructure strategy, including capacity planning, cost optimization, and disaster recovery across AWS Cloud.
- Partner with client stakeholders to define technical roadmaps, translating business requirements into scalable architecture decisions.
- Establish observability standards using CloudWatch, Prometheus, New Relic, and Grafana to proactively detect and resolve production issues.
- Drive Infrastructure as Code adoption using Terraform, standardizing provisioning across environments and reducing configuration drift.
- Conduct architecture and code reviews to enforce security, performance, and scalability best practices across teams.

Systems Engineer - Developer

May 2021 - Feb 2025

Tata Consultancy Services

Gurugram, Haryana, India

- Developed and deployed scalable microservices architectures with seamless RESTful API integrations.
- Built cloud-native applications on AWS and Azure with a focus on performance, security, and cost optimization.
- Implemented Infrastructure as Code practices for automated provisioning and configuration management.
- Built a Kafka-based notification flow for publish events that decoupled the publish pipeline from downstream consumers, reducing service-to-service load and preventing database and server overload during peak publishing.
- Built reactive APIs to efficiently stream and process millions of records at scale.
- Remediated 50+ backend security vulnerabilities, passing security audits with zero critical findings.

- Improved team velocity by approximately 20% by championing GitHub Copilot for code generation, refactoring, and test scaffolding.
- Collaborated with cross-functional teams to deliver high-impact projects that met strict SLA requirements.
- Maintained comprehensive technical documentation and participated in code reviews to ensure quality standards.

PHP Developer

Jan 2021 - Apr 2021

SIB Infotech

Delhi, India

- Developed web applications with an emphasis on performance optimization and database schema design.
- Designed responsive websites and implemented enhancements for existing applications.

KEY PROJECTS

- **Cloud Migration:** Led migration of legacy applications to AWS, reducing infrastructure costs by 35% and improving system performance by 50%.
- **Microservices Platform:** Designed, built, and maintained multiple microservices supporting core business functionality, including a high-availability architecture using Symfony MVC, PHP, Java, SpringBoot, PostgreSQL, and Docker with automated CI/CD for zero-downtime updates—improving modularity, scalability, and maintainability across the platform.
- **CI/CD Migration:** Migrated legacy GoCD pipelines to GitHub Actions, streamlining and standardizing build and deployment workflows across teams.
- **Database Upgrade:** Performed a PostgreSQL version upgrade from 13 to 14 across environments, ensuring data integrity and minimal downtime while improving performance and stability.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (SAA-C03)
- AWS Certified AI Practitioner (AIF-C01)
- AWS Certified Cloud Practitioner (CLF-C02)
- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Google Cloud Compute Basics
- REST API (Intermediate) Skills Certification

EDUCATION

Master of Computer Applications (MCA)

2018 - 2020

TIT Technocrats Group of Institutions, Bhopal, India

Bachelor of Computer Applications (BCA)

2014 - 2017

T.N.B. College, Bhagalpur, India